Independent Study Report

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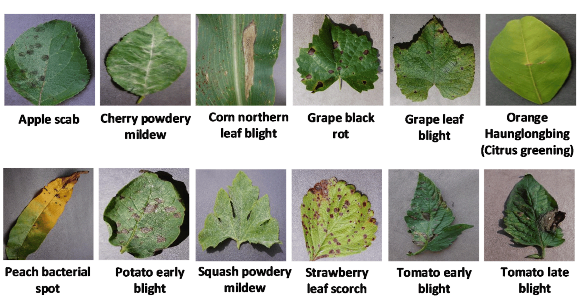
Introduction

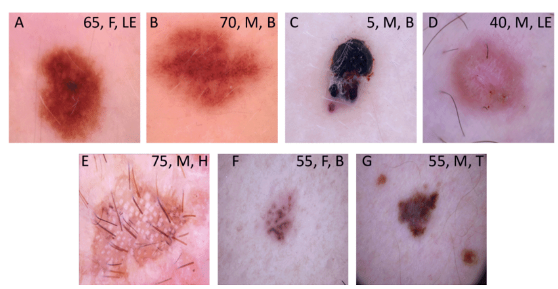
For the Fall 2023 semester, I had an opportunity to explore different Natural Language Processing (NLP) models and compare their performance in a given task. After carefully searching for a good research topic, I decided to work on Cyberbullying detection in SNS comments utilizing different NLP models. The ubiquity and complexity of the language data strongly motivated me to study NLP at a deeper level. This report will address the steps and processes I took this semester while conducting the NLP research.

Doctoral student Ruohan Zong has been my mentor this semester. We had meetings every week to keep track of the research and contacted through email in between the meetings when needed. Throughout the semester, Ruohan was a huge help to me. She not only mentored me in this NLP research project but also provided me with a lot of good feedback on the paper I was previously working on which eventually got admitted to IEEE Big Data 2023 at the end of the semester. I would like to take this opportunity to sincerely thank her.

Search for Research Topic

Early in the semester, we took a considerable amount of time to select the appropriate research topic related to ML/AI for social goods. Topics we considered were Crop Disease Classification, Medical Image Analysis for Disease Diagnosis, and Robbery/Crime Detection. These topics are mostly image learning-based projects where a level of professionalism is required. While there was sufficient image data and previous related research, limitations of professional knowledge prevented me from choosing those topics.





(Examples of image datasets)

Since I was familiar with image learning and classification, I had limited the search scope to projects related to such. However, after realizing that this independent study could be an opportunity for me to learn new techniques such as NLP, I decided to look for text data. Therefore, I chose **cyberbullying detection** for the topic of research.

Cyberbullying Detection Using Different NLP Models

Studying NLP Models

NLP was a completely new topic for me; therefore, an extensive amount of research was required. After exploring related papers, I found that there are different models that used various approaches to language processing. Three models that caught my attention were BERT, RoBERTa, and XLNet. These models became the main topic for the research: comparing their performance in cyberbullying detection.

BERT vs RoBERTa vs XLNet Comparisons

Feature/Aspect	**BERT (Bidirectional Encoder Representations from Transformers)**	**RoBERTa (A Robustly Optimized BERT Pretraining Approach)**	**XLNet (Generalized Autoregressive Pretraining for Language Understanding)**
Developer	Google AI	Facebook AI	Google/CMU
Release Year	2018	2019	2019
Architecture	Transformer-based (bidirectional)	Transformer-based (optimized BERT architecture)	Transformer-based (permutes the order of input tokens)
Training Data	BooksCorpus, English Wikipedia	Extended dataset (including BooksCorpus, CC-News, OpenWebText, Stories)	Larger dataset than BERT (including BooksCorpus, Giga5, ClueWeb)

Key Innovation	Contextualized word representations using bidirectional context	Longer training with more data and optimized hyperparameters	Combines the best of autoregressive and autoencoding approaches
Tokenization	WordPiece	Byte Pair Encoding (BPE)	SentencePiece
Model Size	Available in various sizes (e.g., BERT-base, BERT-large)	Similar to BERT (base and large models)	Available in various sizes (e.g., XLNet-base, XLNet-large)
Performance	High performance on a wide range of NLP tasks	Outperforms BERT on many benchmarks	Outperforms BERT and RoBERTa on several tasks
Application Use	Wide range of NLP tasks (e.g., question answering, NER)	Similar to BERT, particularly effective in fewer data scenarios	High performance in tasks requiring understanding of long context

Literature review

Cyberbullying Detection

For the literature review, I searched if there were attempts to detect cyberbullying utilizing NLP models. A notable contribution in this field is the study by Khan et al. (2022), which emphasizes the use of deep learning models to detect aggressive behavior in social media texts. This research highlights a recurring challenge, the issue of data imbalance, common in datasets where instances of cyberbullying are outnumbered by non-bullying instances. This imbalance can lead to a predictive bias in machine learning models, skewing them towards the majority class. To counter this, various researchers have employed data augmentation and resampling techniques, ensuring a more balanced representation and consequently, more accurate detection.

Comparison of BERT, RoBERTa, and XLNet

Interestingly, there was research that compared these three exact same models with similar implications. Rajapaksha, Farahbakhsh, and Crespi (2021) conducted a study to determine the most effective Transfer Learning model among BERT, XLNet, and RoBERTa for detecting clickbaits in social media. This research is particularly significant as it delves into the comparative strengths and weaknesses of each model in a specific application - clickbait detection.

Method

Training Dataset

To detect cyberbullying, a related dataset with texts and labels was needed. In Kaggle, there was a binary dataset named *aggression_parsed_dataset, twitter_sexism_parsed_dataset, and twitter_racism_parsed_dataset*. Each dataset had binary labels for a large number of internet comments showing whether the comment was aggressive, sexist, or racist. I started with the aggression dataset since it was the most general. However, as mentioned in the literature review, the issue was that the data was imbalanced. It is well established that this could lead to significant challenges in training a machine learning model. When dealing with imbalanced datasets, the model might become biased towards the majority class which was "not aggressive" in this case. This bias could result in a high accuracy rate during training and testing but poor performance in real-world scenarios, especially in correctly identifying truly aggressive comments. The model's inability to effectively recognize and categorize aggressive behavior, due to the disproportionate representation of classes in the training data, would undermine the goal of accurately detecting cyberbullying. In order to handle the problem, I decided to **under-sample** the majority class, the "not aggressive" class, balancing the training dataset.

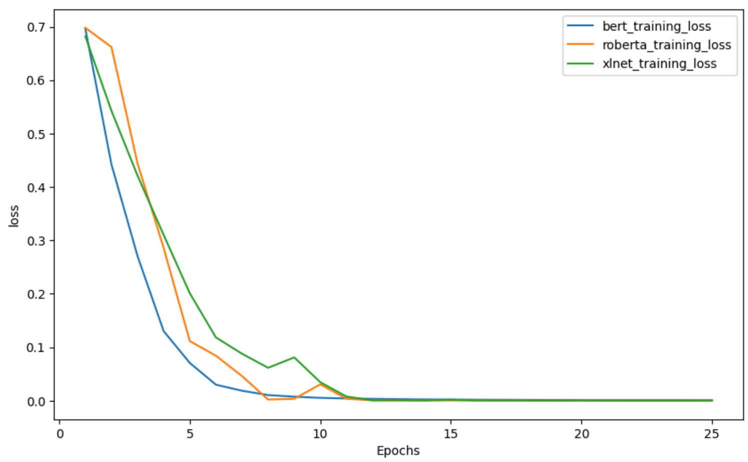
Results

Testing Results using Aggression-parsed-dataset

	BERT	RoBERTa	XLNet
Best Validation Acc	0.84	0.92	0.87

The results of the training were as expected. Considering that RoBERTa improves upon BERT by training on more data, removing the Next Sentence Prediction task, using dynamic masking, and optimizing key hyperparameters, it was expected that RoBERTa would have higher accuracy than BERT.

Loss comparison for BERT, RoBERTa, and XLNet



RoBERTa's training loss (solid red line) decreases steadily, which is a good indicator of learning. Its validation loss (dashed red line) also decreases and exhibits less volatility than XLNet's validation loss, suggesting that RoBERTa is generalizing better. XLNet's training loss (solid green line) is comparable to RoBERTa's, but its validation loss (dashed green line) shows greater fluctuations and, particularly towards later epochs, trends higher than RoBERTa's. This indicates potential issues with XLNet's generalization. I am still trying to identify what caused this fluctuation.

Further Research

Even after this semester, I would like to continue this research project. I have found a lot of room for improvement and modification for this project throughout this semester.

Limitations in Computational Power

Due to limited computational power, I was unable to increase the batch size. Although there was a large amount of data available, only a portion of it could actually be processed. If possible, I would like to take more time and use a better computer to remove these limitations.

Transfer Learning (Racism and Sexism Detection)

The next probable step for the research can be the comparison of these models in transfer learning. Since there is existing but slightly different research (Rajapaksha, Farahbakhsh, and Crespi, 2021) as mentioned in the literature review, it will be great to continue this research and explore how the results are different.

Modification in handling Data Imbalance

While I used simple under-sampling by making the training dataset have the same number of majority classes and minority classes to handle the data imbalance, I would like to try different methods. This approach is only effective when the majority of class data is uniform. However, considering that online comments could be extremely diverse, this is not the case. If I can cluster the majority class, I will be able to sample data from each cluster and under-sample the majority class more representatively.

Conclusion for Independent Study

This semester's independent study was extremely beneficial to me. Considering the recent importance and rise of NLP, it was a great opportunity to explore and dive deeper into this whole new field. I obtained a significant amount of knowledge about general NLP technology. In addition to that, I am now capable of using three different models(BERT, RoBERTa, and XLNet) deeply aware of their characteristics and differences.

I initially wanted to write a short paper that I could submit to undergraduate journals during this independent study. However, as I continued this research, I found it more interesting and wanted to try writing a decent high-quality paper regarding this topic. I will continue this research about Cyberbullying detection even after this semester ends.

Works Cited

Rajapaksha, P., Farahbakhsh, R., & Crespi, N. (2021). BERT, XLNet or RoBERTA: the best transfer learning model to detect clickbaits. *IEEE Access*, *9*, 154704–154716. https://doi.org/10.1109/access.2021.3128742

Khan, U., Khan, S., Rizwan, A., Atteia, G., Jamjoom, M., & Samee, N. A. (2022). Aggression Detection in Social Media from Textual Data Using Deep Learning Models. *Applied Sciences*, *12*(10), 5083. https://doi.org/10.3390/app12105083